INSTITUTE OF INFORMATION TECHNOLOGY & MANAGEMENT

QUESTION BANK SOLUTIONS

Subject: Internet of Things (IoT) | Subject Code: BCA-306 | BCA SEM VI

Unit 1 & Unit 2 — Complete Solutions

UNIT 1 — IoT: Vision, M2M, Design Principles & Communication

Q.1) Short Notes — Unit 1 Topics	**10 Marks** *[CO1]*

(a) Vision of IoT

The vision of IoT is a world where every physical object is embedded with sensors, connectivity, and intelligence, enabling autonomous communication and cooperation without human intervention.

- **Every device senses, collects, and transmits data over the internet autonomously.**Smart Connectivity:
- **Intelligence embedded invisibly in everyday objects — home appliances, vehicles, industrial machines.**Ubiquitous Computing:
- **Systems make autonomous decisions, reducing manual effort and improving resource utilisation.**Automation & Efficiency:
- **Massive sensor data enables predictive analytics, pattern recognition, and smarter services.**Data-Driven Insight:
- **Smart homes, healthcare monitoring, smart cities, precision agriculture, and assisted living.**Improved Quality of Life:

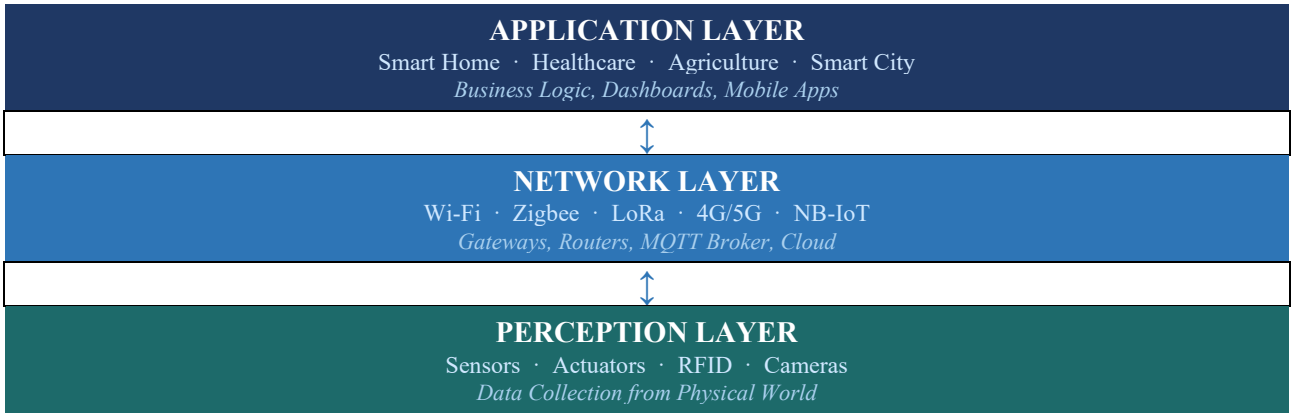

Figure 1: IoT Three-Layer Architecture — Perception, Network, and Application Layers

(b) Conceptual Framework of IoT

The IoT conceptual framework defines how IoT devices, networks, services, and applications are organised and interact. It comprises three core layers — Perception (data collection), Network (data transmission), and Application (data consumption/service delivery) — as illustrated in Figure 1 above.

- **Sensors, actuators, RFID tags, cameras. Generates raw data from the physical world.**Perception Layer:
- **Wi-Fi, Zigbee, Bluetooth, LoRa, 4G/5G. Routes data to cloud or edge processing.**Network Layer:
- **Smart home apps, industrial dashboards, healthcare platforms. Delivers actionable services.**Application Layer:

(c) M2M Communication

Machine-to-Machine (M2M) communication is the direct exchange of data between devices over wired or wireless networks without human intervention.

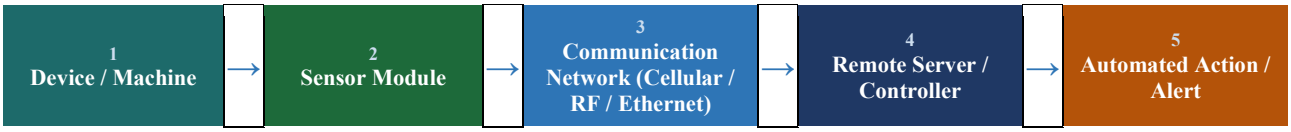

Figure 2: M2M Communication Flow — Device → Sensor → Network → Server → Automated Action

- **Cellular (2G/3G/4G), Zigbee, Bluetooth, Wi-Fi, satellite, wired Ethernet.**Technologies:
- **M2M is point-to-point between specific machines. IoT is broader — connects heterogeneous devices, services, and humans through cloud platforms.**M2M vs IoT:

(d) Sources of IoT

- **Temperature, humidity, pressure, motion, light — the primary source of real-world IoT data.**Physical Sensors:
- **Attached to objects for identity and location tracking in supply chain, retail, and healthcare.**RFID Tags:
- **GPS, accelerometers, gyroscopes, cameras — rich personal context data.**Smartphones & Wearables:
- **CNC machines, robots, conveyor systems reporting operational metrics and fault codes.**Industrial Machines:
- **Smart electricity/gas/water meters; traffic cameras; environmental monitoring stations.**Smart Infrastructure:
- **Participatory sensing apps contributing to large-scale environmental datasets.**Social & Crowd-sourced Data:

(e) IoT Examples in Real Life

- **Google Nest thermostats, Philips Hue smart lighting, Amazon Echo — devices that automate household tasks.**Smart Home:
- **Wearable ECG monitors, insulin pumps with auto-dose adjustment, remote patient monitoring systems.**Smart Healthcare:
- **Soil moisture sensors, drone-based crop monitoring, automated drip irrigation systems.**Smart Agriculture:
- **GPS fleet management, connected cars sharing traffic data, smart parking systems.**Smart Transportation:
- **Predictive maintenance detecting bearing wear, automated quality control using computer vision.**Industrial IoT (IIoT):

Q.2) IoT — Definition, Vision, Sources & Technology	**10 Marks** *[CO1]*

Definition of IoT

Internet of Things (IoT) is a global network of physical objects embedded with sensors, software, and connectivity technologies that enable them to collect, exchange, and act on data over the internet — extending internet connectivity beyond computers and smartphones to every object in the physical world.

IoT/M2M System Layers (Architectural View)

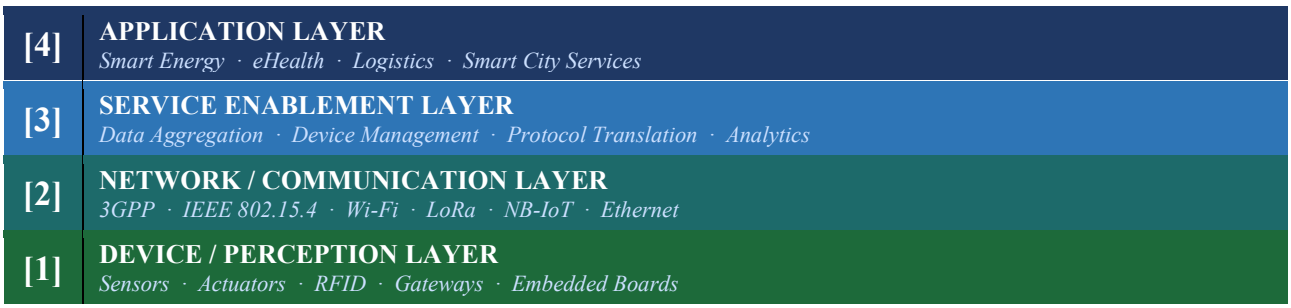

Figure 3: IoT/M2M System Layers — from Device Layer to Application Layer

Technology Behind IoT

- **Arduino, Raspberry Pi, ARM Cortex — low-power computing at the edge.**Embedded Systems & Microcontrollers:
- **Wi-Fi, Bluetooth, Zigbee, Z-Wave, LoRa, NB-IoT, 5G — each suited to different range/power trade-offs.**Communication Protocols:
- **AWS IoT, Azure IoT Hub, Google Cloud IoT — data storage, processing, and analytics at scale.**Cloud Computing:
- **MQTT, CoAP, AMQP — lightweight messaging protocols for device-to-cloud communication.**Middleware Platforms:
- **Analysing sensor data streams for anomaly detection, predictive maintenance, and automation.**AI & ML:
- **TLS/SSL encryption, PKI authentication, secure boot, hardware security modules (HSM).**Security:

— OR —

IoT/M2M Layers & Design Standardisation — Differences between IoT and M2M

The four-layer IoT/M2M architecture is shown in Figure 3. Standardisation ensures interoperability across heterogeneous vendors and platforms. Key standards: oneM2M (global IoT), ETSI M2M, IEEE 802.15.4, 3GPP (NB-IoT/LTE-M).

IoT vs M2M	IoT
Communication Model	Broad: connects devices, services & humans via cloud
Intelligence	AI-enabled, context-aware decisions
Network	Internet + cloud platforms
Scale	Billions of diverse devices
Standards	oneM2M, W3C WoT, OPC-UA
Examples	Smart city, precision farming, IIoT

Q.3) M2M Communication — Role in IoT, Sources of IoT	**10 Marks** *[CO1]*

M2M Communication — Working Principle

M2M is the automated exchange of data between machines via wired or wireless networks without human involvement. The communication flow is shown in Figure 2 (above).

- **A sensor/device monitors a physical parameter (temperature, flow, position).**Step 1 — Data Collection:
- **Data is sent via a communication channel (cellular, RF, Ethernet).**Step 2 — Transmission:
- **A remote server or controller receives the data.**Step 3 — Reception:
- **System processes data and may send a control command back (e.g., close a valve, send an alert).**Step 4 — Automated Action:

Role of M2M in IoT

- Forms the communication backbone for IoT device interactions.
- Enables remote monitoring, diagnostics, and control of deployed devices.
- Supports real-time alerting and automated responses in industrial and utility systems.
- Provides the data streams that IoT analytics platforms consume.

— OR —

Sources of IoT — Detailed Description

- **Measure temperature, humidity, pressure, motion, light, air quality — largest source of IoT data.**Physical Environment Sensors:
- **Passive and active tags on products, pallets, animals, and people providing identity and location data.**RFID Systems:
- **GPS, accelerometers, cameras, microphones — rich personal context data.**Smartphones & Mobile Devices:

- **CNC machines, turbines, conveyor belts reporting operational metrics and fault codes.**Industrial Machines (IIoT):
- **Smart meters (electricity/gas/water), traffic cameras, environmental monitoring stations.**Smart Infrastructure:
- **Participatory sensing through smartphone apps contributing large-scale city and environmental data.**Social & Crowd-sourced Data:
- **GPS constellations and weather satellites providing geospatial context.**Satellite & Remote Sensing:

Q.4) IoT/M2M Layers, Standardisation & Communication Technologies	**10 Marks** *[CO1, CO2]*

Communication Technologies Comparison

Technology | Range | Power | Use Case	Key Properties
Wi-Fi 802.11ac | ~50–100 m | High | Home IoT, high-data streaming	High bandwidth (up to 1 Gbps); needs mains power; easy to deploy
Bluetooth / BLE | ~10–100 m | Low | Wearables, health monitors, peripherals	Star topology; BLE operates on 2.4 GHz with 3 µW–15 mW power
Zigbee 802.15.4 | ~10–100 m | Very Low | Smart home mesh, industrial WSN	Mesh capable; IEEE 802.15.4 PHY/MAC; 250 kbps; 10-year battery life
LoRa / LoRaWAN | 2–15 km | Very Low | Agriculture, smart city, asset tracking	Sub-GHz ISM band; chirp spread spectrum; years on coin cell battery
NB-IoT (LTE-M) | ~1–10 km | Very Low | Smart meters, remote sensors	Licensed spectrum; excellent indoor penetration; < 1 mA avg. current
5G Cellular | Wide area | Medium | Autonomous vehicles, industrial automation	< 1 ms latency; network slicing; massive IoT (mMTC) mode for devices

— OR —

Why Standardisation is Important

- **Devices from different manufacturers can communicate over standard APIs and protocols.**Interoperability:
- **Common libraries, SDKs, and certification programmes reduce development effort.**Cost Reduction:
- **Standardised security frameworks (TLS, DTLS, OSCAR) ensure baseline protection.**Security & Compliance:
- **Standardised protocols allow seamless addition of new device types without redesign.**Scalability:

Q.5) Design Principles for Connected Devices	**10 Marks** *[CO2]*

Design Principles Overview

Connected IoT devices must be designed around five key principles: layered architecture, data enrichment, data consolidation, ease of design, and affordability.

- **Adding contextual metadata to raw sensor readings — timestamps, device IDs, geo-coordinates, calibration factors, and external data fusion (weather APIs, map services) to make raw values actionable.**Data Enrichment:
- **Aggregating high-frequency sensor data at the edge (gateway) before cloud upload. Includes time-window averaging, deduplication, and schema harmonisation across heterogeneous devices.**Data Consolidation:
- **Modular hardware (Arduino shields, Raspberry Pi HATs), standard APIs (REST, MQTT), visual programming tools (Node-RED), and cloud managed services (AWS IoT Core) enable rapid prototyping.**Ease of Designing:
- **ESP8266 ~$1, Arduino Nano ~$3, ESP32 ~$5 — low-cost SoCs make large-scale deployment economically viable. Open-source hardware and pay-as-you-go cloud pricing reduce total cost of ownership.**Affordability:

- **Using standard protocols (MQTT, CoAP, OPC-UA) ensures devices from different vendors work together and simplifies system integration.**Standardisation:

UNIT 2 — Hardware for IoT: Sensors, RFID, WSN & Embedded Platforms

Q.6) Short Notes — Unit 2 Hardware Topics	**10 Marks** *[CO2, CO3]*

(a) Digital Sensors

A digital sensor outputs a discrete digital signal via a communication protocol (I²C, SPI, UART, 1-Wire) rather than a continuous analogue voltage. The sensor internally digitises the physical measurement.

Figure 4: Sensor Working Principle — Physical Stimulus → Transduction → Signal Conditioning → Digital Output → Microcontroller

- **DHT22 (temperature/humidity), DS18B20 (1-Wire temperature), BMP280 (I²C pressure), HC-SR04 (ultrasonic distance), PIR (motion — HIGH/LOW).**Examples:
- **Noise-immune; direct microcontroller interfacing without external ADC; often factory-calibrated.**Advantage over analogue:

(b) Actuators

An actuator converts an electrical control signal from a microcontroller into a physical action — motion, heat, light, sound, or fluid flow.

- **Solenoids, relays, DC motors, servo motors, stepper motors.**Electrical Actuators:
- **Use pressurised fluid for high-force industrial applications.**Hydraulic Actuators:
- **Use compressed air for factory automation and pneumatic cylinders.**Pneumatic Actuators:
- **Relay modules (appliance control), servo motors (smart locks), solenoid valves (irrigation), buzzers (alarms).**IoT Actuators:

(c) Wireless Sensor Network (WSN)

A WSN is a network of spatially distributed sensor nodes that wirelessly monitor physical conditions and report to a central system.

Figure 5: WSN Sensor Node Architecture — Sensing Unit, Processing Unit, Transceiver, and Power Unit

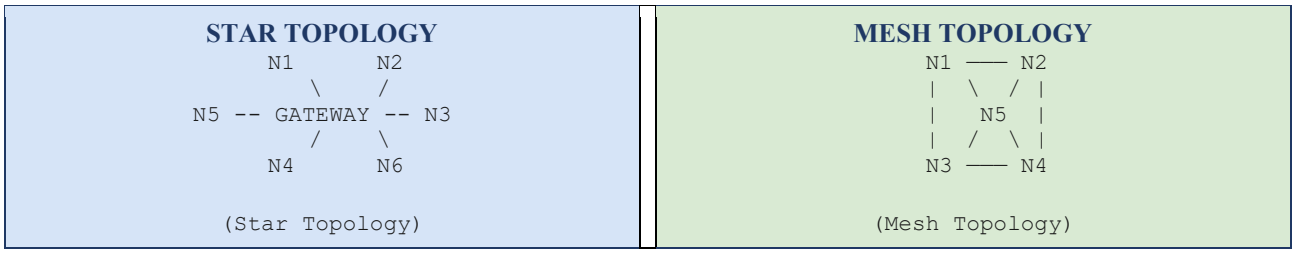

Figure 6: WSN Topologies — Star Topology (centralised) and Mesh Topology (self-healing)

- **Zigbee (IEEE 802.15.4), Bluetooth Mesh, LoRa, 6LoWPAN.**Protocols:
- **Environmental monitoring, precision agriculture, structural health monitoring, smart buildings.**Applications:

(d) Participatory Sensing Technology

Participatory sensing is an approach where citizens contribute sensor data using smartphones to create large-scale collaborative IoT datasets.

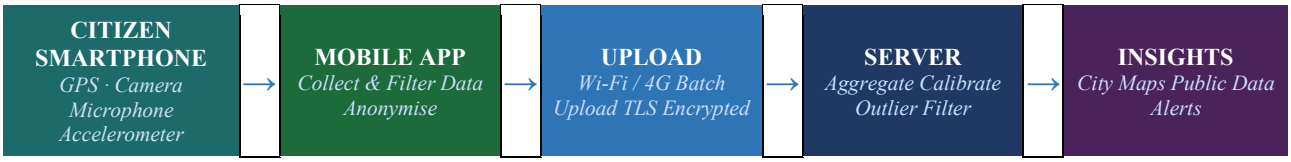

Figure 7: Participatory Sensing Data Flow — Citizen Smartphone → App → Upload → Server → Insights

- **OpenSignal (mobile coverage), PurpleAir (air quality), Waze (traffic), iNaturalist (biodiversity).**Examples:
- **Data quality variability, privacy concerns (GPS traces), irregular participation.**Challenges:

(e) Architecture of Arduino Board

Figure 8: Arduino Uno Board Architecture — Digital I/O, ATmega328P Core, Analogue Input Pins

- **8-bit AVR, 16 MHz, 32 KB Flash, 2 KB SRAM, 1 KB EEPROM.**ATmega328P:
- **14 digital pins (6 PWM), 6 analogue pins (10-bit ADC).**I/O:
- **UART, SPI, I²C interfaces; USB-to-serial via ATmega16U2.**Communication:
- **USB 5V or barrel jack 7–12V; onboard 3.3V and 5V regulators.**Power:

Q.7) Sensors — Working Principle, Types, Analogue vs Digital	**10 Marks** *[CO2]*

What is a Sensor?

A sensor is a device that detects and measures a physical quantity from the environment and converts it into an electrical signal readable by a microcontroller or IoT gateway. The conversion process is called transduction.

Figure 9: Sensor Signal Chain — Physical Input → Transducer → ADC/Conditioning → Digital Output

Five Types of Sensors Used in IoT

- **Thermistor/thermocouple changes resistance/voltage with temperature. Used in smart HVAC, cold-chain, patient monitoring.**1. Temperature Sensor (DHT22, LM35, DS18B20):
- **Capacitive polymer film changes capacitance as moisture varies. Used in greenhouses, HVAC, weather stations.**2. Humidity Sensor (DHT11, SHT31):

- **Pyroelectric sensor generates voltage pulse when it detects infrared radiation from warm bodies. Used in security systems, automatic lighting.**3. Motion/PIR Sensor (HC-SR501):
- **Emits 40 kHz pulse, measures time-of-flight of echo. Distance = (time × 343 m/s) / 2. Used in parking sensors, liquid level measurement.**4. Ultrasonic Distance Sensor (HC-SR04):
- **Photoresistor changes resistance with light intensity. Used in smart street lighting, display brightness adjustment.**5. Light Sensor (LDR, BH1750):

— OR —

Analogue vs Digital Sensors

Aspect	Analogue Sensor
Output Signal	Continuous voltage/current proportional to measured value
Interfacing	Requires external ADC to connect to microcontroller
Noise Sensitivity	Susceptible to electromagnetic noise over long wires
Memory/Processing	Simpler hardware; complex signal conditioning needed
Examples	LM35, LDR, Potentiometer, Thermocouple

Q.8) RFID Technology & WSN/Actuators	**10 Marks** *[CO2]*

RFID System Components

Figure 10: RFID System — Tag ↔ Reader → Middleware → Backend System/Database

Types of RFID Tags

Figure 11: RFID Tag Types — Passive, Active, and Semi-Passive Comparison

- **LF (125 kHz, ~10 cm — animal tracking), HF (13.56 MHz, ~1 m — smart cards/NFC), UHF (860–960 MHz, ~10 m — retail/supply chain), Microwave (2.45 GHz — industrial).**RFID Frequency Bands:
- **Supply chain inventory, patient wristbands (healthcare), smart access control, livestock management, asset tracking.**IoT Applications of RFID:

— OR —

(a) Actuators in IoT

- **Converts electrical energy to continuous rotational motion. Used in conveyor belts, fans, pumps.**DC Motor + Driver (L298N):
- **PWM-controlled precise angle positioning. Used in robotic arms, smart door locks, camera gimbals.**Servo Motor:
- **Discrete step-angle control without feedback. Used in 3D printers, CNC machines, valve actuators.**Stepper Motor:
- **Electromagnetic linear actuator for fluid/gas flow control. Used in irrigation, pneumatics, door bolts.**Solenoid Valve:

- **Electrically controlled switch — isolates high-voltage loads from 5V microcontroller. Core component of smart home automation.**Relay Module:

(b) Wireless Sensor Network (WSN)

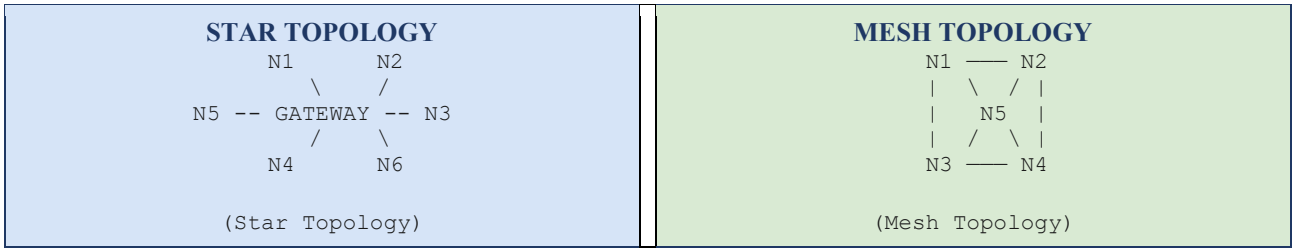

Figure 12: WSN Topologies — Star (centralised, simple routing) vs Mesh (distributed, fault-tolerant)

Figure 13: WSN Node Internal Architecture — all components on a single sensor node

- **WirelessHART (industrial), Zigbee PRO (commercial), 6LoWPAN (IPv6 over constrained networks), LoRaWAN (wide-area WSN).**Protocols:
- **Forest fire detection, structural health monitoring, battlefield surveillance, smart agriculture, indoor positioning.**Applications:
- **Battery lifetime, limited bandwidth, node failure, RF interference, security vulnerabilities.**Challenges:

Q.9) Embedded Platforms for IoT	**10 Marks** *[CO3, CO4]*

Embedded Computing Basics

Embedded systems are specialised computing systems with dedicated functions, real-time constraints, and resource limitations (power, memory, processing). IoT embedded platforms must balance these constraints with connectivity and cost.

Platform	Core	RAM	OS	Networking	Best IoT Role
Arduino Uno	ATmega328P 8-bit 16MHz	2 KB	None (Bare Metal)	None (Shields needed)	Sensor/Actuator Control
NetArduino	ARM926EJ-S 32-bit 400MHz	8 MB	.NET Micro Framework	Built-in Ethernet	Networked Embedded IoT
Raspberry Pi 4	Cortex-A72 64-bit 1.5GHz	4 GB	Linux (Debian/Ubuntu)	Wi-Fi + BT + Ethernet	Edge AI, Gateway, Media Processing
Beagle Bone	Cortex-A8 32-bit 1GHz	512 MB	Linux + PRU Real-time	100M Ethernet + USB	Industrial RT Control, Robotics
Intel Galileo	Intel Quark 32-bit 400MHz	256 MB	Yocto Linux	Ethernet (Arduino pins)	Arduino + Linux Bridge Platform
ARM Cortex-M	Cortex-M4/M7 up to 480MHz	< 1 MB	Bare/FreeRTOS	BLE / Wi-Fi (varies by board)	Ultra-low Power Sensor Nodes

Figure 14: IoT Embedded Platform Comparison — Arduino, NetArduino, Raspberry Pi, Beagle Bone, Intel Galileo, ARM Cortex

Platform Selection Guide

- **→ Arduino Uno or ARM Cortex-M based board.**Simple sensor/actuator control:
- **→ NetArduino.**Networked IoT without full OS:
- **→ Raspberry Pi 4.**Edge AI, video, full web server:

- → **Beagle Bone Black (PRU cores).**Hard real-time Linux I/O:
- → **Intel Galileo.**Arduino code + Linux extension:
- → **ARM Cortex-M (STM32, nRF52840).**Ultra-low-power wearables or sensors:

— OR —

Raspberry Pi 4 for IoT

- **Broadcom BCM2711 — ARM Cortex-A72 quad-core, 1.5 GHz (64-bit).**Processor:
- **2/4/8 GB LPDDR4 RAM; MicroSD card storage.**Memory:
- **Gigabit Ethernet, Wi-Fi 802.11ac (dual-band), Bluetooth 5.0.**Connectivity:
- **40-pin GPIO (UART, SPI, I²C, PWM), 2× USB 3.0, 2× USB 2.0, 2× micro-HDMI.**I/O:
- **Smart home hub (Home Assistant), industrial gateway (MQTT broker), computer vision node (OpenCV + TFLite), environmental monitoring.**IoT Applications:

Q.10) NetArduino vs Arduino & Participatory Sensing	**10 Marks** *[CO3]*

NetArduino vs Arduino — Visual Comparison

ARDUINO UNO	NETARDUINO
• 8-bit AVR @ 16 MHz	• 32-bit ARM @ 400 MHz
• 32 KB Flash / 2 KB SRAM	• 16 MB Flash / 8 MB RAM
• No built-in networking	• Built-in 10/100 Ethernet
• C/C++ (Arduino IDE)	• C# / .NET Micro Framework
• No OS — bare metal	• Soft real-time threading
• Best: Real-time sensor/actuator	• Best: Networked IoT apps

Figure 15: NetArduino vs Arduino — Key Differences in Architecture, Networking, and IoT Suitability

Feature	Arduino Uno
Processor	ATmega328P — 8-bit 16 MHz
RAM	2 KB SRAM
Networking	None built-in (shields needed)
Programming	C/C++ (Arduino IDE)
Real-time	Bare-metal — no OS overhead
Cost	~$3–$10
Best For	Real-time sensor/actuator tasks

- **Built-in Ethernet eliminates shields; C# and .NET enable enterprise-grade code; 8 MB RAM supports JSON parsing, HTTP, and MQTT clients; Arduino-compatible pinout reuses existing shields.**Why NetArduino suits IoT:

— OR —

Participatory Sensing Technology

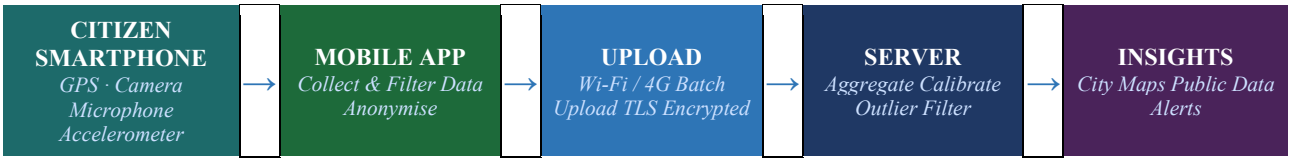

Figure 16: Participatory Sensing Flow — Citizen devices contribute to large-scale IoT datasets

- **Citizen PM2.5 sensors map pollution at street level across entire cities.**Air Quality (PurpleAir, Aclima):
- **140 million users provide real-time speed and incident data.**Traffic (Waze):
- **Smartphone accelerometers detect seismic activity globally.**Earthquake (MyShake):

- **k-anonymity, differential privacy, on-device ML, explicit opt-in, GDPR compliance.**Privacy Solutions:

Q.11) Smart Home Automation IoT System Design	**10 Marks** *[CO1, CO2, CO3]*

System Architecture

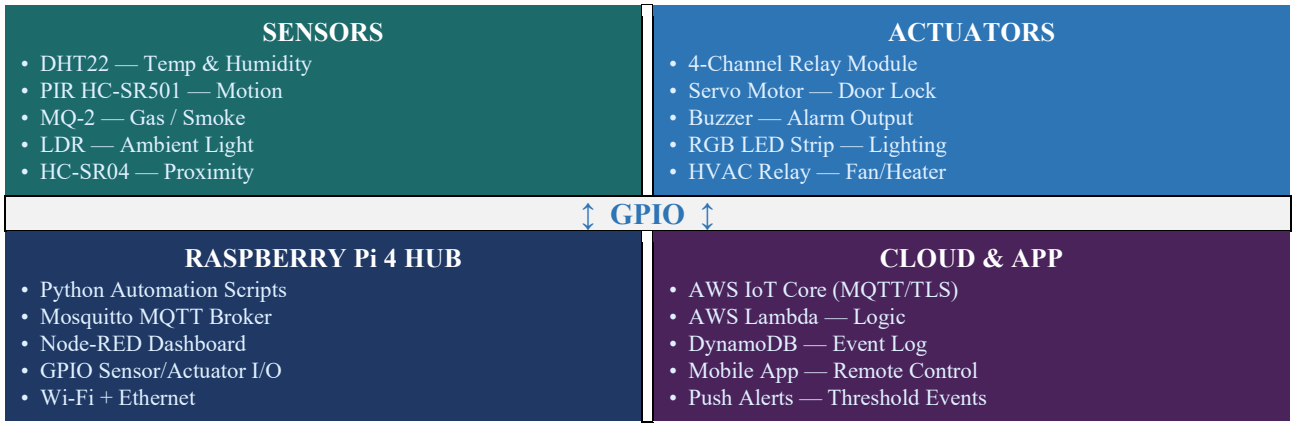

Figure 17: Smart Home IoT Architecture — Sensors, Raspberry Pi Hub, Actuators, and Cloud/App Layer

Sensors & Actuators

- **Digital temperature and humidity — room climate monitoring for HVAC control.**DHT22:
- **Motion detection — security alerts and automatic lighting trigger.**PIR HC-SR501:
- **Smoke and combustible gas detection — fire alarm relay.**MQ-2 Gas Sensor:
- **Controls high-voltage appliances (fan, heater, lights) from 5V GPIO.**4-Channel Relay Module:
- **Smart door lock — precise angular control for bolt mechanism.**Servo Motor:

Embedded Platform — Raspberry Pi 4

RPi 4 selected as central hub: runs Linux, Python automation, Mosquitto MQTT broker, and Node-RED dashboard. GPIO pins interface directly with all sensors and relay modules. Wi-Fi provides wireless connectivity to AWS IoT Core.

Data Flow

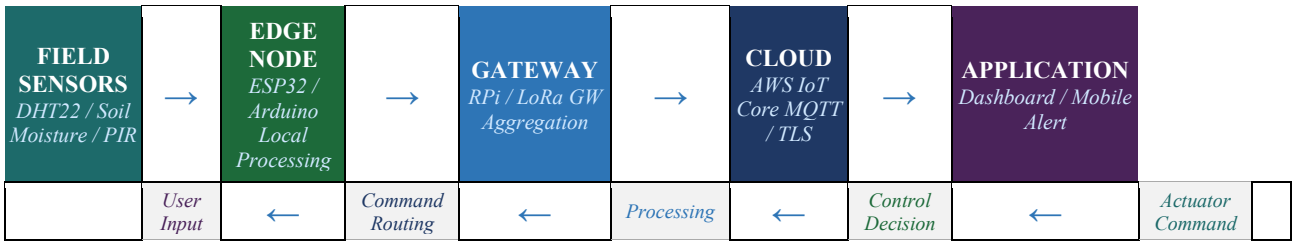

Figure 18: End-to-End IoT Data Flow — Field Sensors → Edge Node → Gateway → Cloud → Application (and return command path)

- Sensor → RPi GPIO → MQTT publish → AWS IoT Core → Lambda → DynamoDB → Mobile App → Command back via MQTT → Relay/Servo actuated.

Q.12) WSN vs RFID & End-to-End IoT Data Flow	**10 Marks** *[CO2]*

WSN vs RFID Comparison

Parameter	WSN

Sensing Capability	Measures physical quantities — temperature, pressure, motion
Range	10 m to several km (LoRa WSN)
Power	Battery with duty-cycling; years of life
Cost per Node/Tag	$5 – $30 per node
Networking	Multi-hop mesh; self-organising
Identification	Node identified — not individual objects
Best IoT Scenario	Environmental monitoring, smart agriculture, structural health

- **Continuous parameter measurement over a large area is needed with self-healing network coverage.**Prefer WSN when:
- **Identifying and tracking discrete physical objects automatically at very low per-unit cost.**Prefer RFID when:

— OR —

End-to-End IoT Data Flow

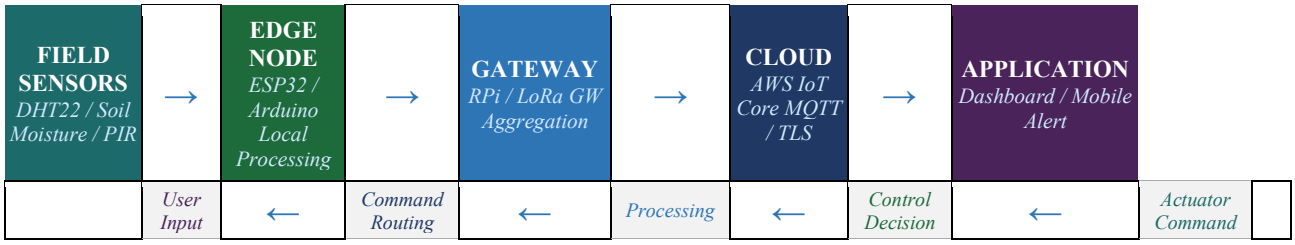

Figure 19: End-to-End IoT Data Flow — Upward data path (sensor to cloud) and downward command path

Stage-by-Stage Explanation

- **ESP32 node wakes from deep sleep, reads sensors (DHT22, soil moisture), packages JSON: {device:"node01", temp:23.5, ts:1710000000}.**Stage 1 — Sensing:
- **If soil_moisture < 30% → trigger relay immediately (edge decision — no cloud dependency).**Stage 2 — Local Processing (Edge):
- **MQTT over Wi-Fi to Raspberry Pi broker. TLS 1.3 + X.509 certificate authentication.**Stage 3 — M2M Transmission:
- **AWS IoT Core → Lambda → stream analytics → InfluxDB time-series database.**Stage 4 — Cloud Processing:
- **Grafana dashboard, mobile push alerts; commands return via MQTT → actuator triggered.**Stage 5 — Application:

Q.13) Short Notes — Mixed Unit 1 & Unit 2	**10 Marks** *[CO1, CO3, CO4]*

(a) Microcontroller vs Microprocessor

Feature	Microcontroller
Integration	CPU + RAM + Flash + I/O on single chip
Memory	On-chip 2 KB – 512 KB RAM
I/O Peripherals	Built-in GPIO, UART, SPI, I²C, ADC, PWM
OS Support	Bare-metal or lightweight RTOS
Power	Very low (µA in sleep)
Speed	8 MHz – 480 MHz

IoT Role	Sensor interfacing, actuator control, edge node

(b) ARM Cortex IoT Platforms

ARM architecture dominates IoT due to its power efficiency and scalable performance range:

- **Ultra-low power, 32-bit. Simple sensors and wearables. Examples: STM32F0, nRF51822.**Cortex-M0/M0+:
- **DSP/FPU extensions. Signal processing, motor control, embedded ML. Examples: STM32F4, Arduino Due.**Cortex-M4/M7:
- **Application processors running Linux. Raspberry Pi 3/4, NXP i.MX. Edge AI, IoT gateways.**Cortex-A53/A72:

(c) IoT Architectural Layers

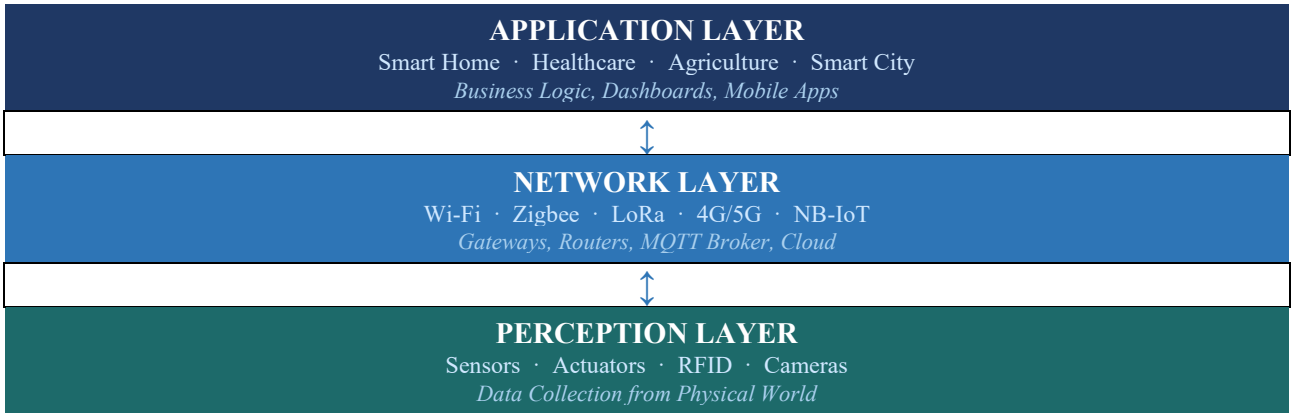

Figure 20: IoT Three-Layer Architecture — Perception → Network → Application

- **Physical world interface — sensors, RFID, cameras, GPS. Data acquisition.**Perception Layer:
- **Transmits data — Wi-Fi, Zigbee, LoRa, 4G/5G. Gateways, routers.**Network Layer:
- **Aggregation, device management, semantic processing — MQTT broker, REST APIs.**Middleware Layer:
- **User services — dashboards, alerts, control panels. Delivers insights to users.**Application Layer:

(d) Beagle Bone in IoT Development

- **Two 200 MHz Programmable Real-time Units allow hard real-time I/O alongside Linux — critical for industrial motor control.**Unique PRU Cores:
- **65+ configurable pins — CAN bus, ADC for analogue sensors, UART for RS-232.**Rich GPIO:
- **Runs Debian — Python, Node.js, ROS (Robot Operating System) for robotics.**Linux OS:
- **Industrial sensor gateway, robotics, CNC control, network-attached instruments.**IoT Use Cases:

(e) Communication Stack in M2M Systems

- **Radio hardware — GSM/LTE module, Zigbee transceiver. Bit-level transmission.**Physical Layer:
- **IEEE 802.15.4, 802.11, 3GPP. Framing, error detection (CRC), MAC addressing.**Data Link Layer:
- **IPv4/IPv6, 6LoWPAN. Routing between device and server.**Network Layer:
- **UDP (low overhead for constrained devices) or TCP (reliable delivery).**Transport Layer:
- **MQTT, CoAP, AMQP, HTTP/REST. Carries the actual sensor data payload.**Application Layer:

Q.14) IoT Conceptual Framework & Embedded System Constraints	**10 Marks** *[CO1, CO3, CO4]*

IoT Conceptual Framework

[4]	**APPLICATION LAYER** *Smart Energy · eHealth · Logistics · Smart City Services*

[3]	**SERVICE ENABLEMENT LAYER** *Data Aggregation · Device Management · Protocol Translation · Analytics*
[2]	**NETWORK / COMMUNICATION LAYER** *3GPP · IEEE 802.15.4 · Wi-Fi · LoRa · NB-IoT · Ethernet*
[1]	**DEVICE / PERCEPTION LAYER** *Sensors · Actuators · RFID · Gateways · Embedded Boards*

Figure 21: IoT Conceptual Framework — Four Layers with responsibilities and examples

Real-World Deployment — Smart Hospital

- **Patient wristbands (pulse oximeter + temperature + accelerometer) + room CO_2/humidity sensors.**Perception:
- **Bluetooth LE → ward gateway → hospital Wi-Fi → AWS IoT Core (TLS-encrypted MQTT).**Network:
- **Hospital IoT platform filters alerts, stores 24-hr vitals history, correlates room conditions.**Middleware:
- **Nurse dashboard with real-time vitals; automated alert to mobile if SpO_2 drops below 95%.**Application:
- **Nurse command from app → MQTT → thermostat actuator adjusts room HVAC set point.**Bidirectional:

— OR —

Embedded System Constraints in IoT

- **Battery-powered nodes must last years. Requires duty-cycling (sleep/wake), interrupt-driven wake-up, and sub-µA sleep MCU (STM32L0: 500 nA stop mode).**Power:
- **Arduino Uno has only 2 KB SRAM — cannot run JSON parsers, TLS stacks, or complex data structures. ESP32 (520 KB) or RPi (4 GB) needed for heavier tasks.**Memory:
- **8–240 MHz range. FPU required for real-time signal processing. ML inference needs dedicated accelerators (Cortex-M55 + Ethos U-series NPU).**Processing:
- **Adding Wi-Fi/Bluetooth increases cost and power. LoRa is ideal for wide-area low-power nodes. BLE for personal area applications.**Connectivity:

Platform	Core	RAM	OS	Networking	Best IoT Role
Arduino Uno	ATmega328P 8-bit 16MHz	2 KB	None (Bare Metal)	None (Shields needed)	Sensor/Actuator Control
NetArduino	ARM926EJ-S 32-bit 400MHz	8 MB	.NET Micro Framework	Built-in Ethernet	Networked Embedded IoT
Raspberry Pi 4	Cortex-A72 64-bit 1.5GHz	4 GB	Linux (Debian/Ubuntu)	Wi-Fi + BT + Ethernet	Edge AI, Gateway, Media Processing
Beagle Bone	Cortex-A8 32-bit 1GHz	512 MB	Linux + PRU Real-time	100M Ethernet + USB	Industrial RT Control, Robotics
Intel Galileo	Intel Quark 32-bit 400MHz	256 MB	Yocto Linux	Ethernet (Arduino pins)	Arduino + Linux Bridge Platform
ARM Cortex-M	Cortex-M4/M7 up to 480MHz	< 1 MB	Bare/FreeRTOS	BLE / Wi-Fi (varies by board)	Ultra-low Power Sensor Nodes

Figure 22: Embedded Platforms Addressing IoT Constraints — from bare-metal MCUs to Linux SBCs

Q.15) Smart Agriculture IoT System (Case Study)	**10 Marks** *[CO1, CO2, CO3, CO4]*

System Architecture Diagram

FIELD SENSORS	→	ESP32 NODE	→	RPi GATEWAY	→	AWS IoT CORE	→	DASHBOARD
Soil Moisture Temp/Humidity Rainfall DS18B20		Read Sensors Local Logic MQTT Publish Deep Sleep		Aggregate Data Local Storage Forward to Cloud Node-RED		MQTT Broker Lambda Timestream DB SageMaker ML		Grafana Charts Mobile Alerts Pump Control Irrigation Map

Figure 23: Smart Agriculture IoT System — Field Sensors → ESP32 Node → RPi Gateway → AWS Cloud → Dashboard

(a) Sensors & Actuators

- **Volumetric water content at root depth (~15 cm). Analogue output to ESP32 ADC.**Soil Moisture Sensor (Capacitive v1.2):
- **Monitors ambient conditions for evapotranspiration rate calculation and irrigation scheduling.**DHT22 (Temp & Humidity):
- **Soil temperature at root zone. 1-Wire digital protocol.**DS18B20 Waterproof:
- **Detects rain events — automatically suspends irrigation.**Rainfall Sensor (Tipping Bucket):
- **12V DC pump and per-zone solenoid valves — controlled by ESP32 GPIO via relay.**Relay Module + Solenoid Valve:

(b) Embedded Platform & Communication

- **Dual-core 240 MHz, Wi-Fi + BLE, built-in ADC, deep sleep to 10 µA. Reads sensors, applies local threshold logic, publishes MQTT.**ESP32 Field Node:
- **Aggregates data from multiple ESP32 nodes, runs Mosquitto MQTT broker, caches data during outages, forwards to AWS IoT Core.**Raspberry Pi 4 Gateway:
- **LoRaWAN for large farms (2–15 km range, years on battery); Wi-Fi for greenhouses; NB-IoT as cellular backup.**Communication:
- **MQTT over TLS. Topics: farm/zone1/soil_moisture, farm/zone1/pump/command.**Protocol:

(c) Data Flow — Field to Cloud

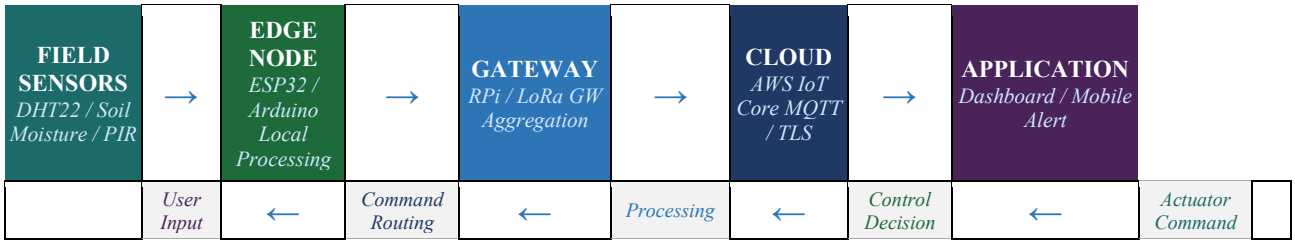

Figure 24: Smart Agriculture Data Flow — Sensor data collection, edge processing, cloud analytics, and return control commands

- **ESP32 wakes from deep sleep every 15 min, reads all sensors, packages JSON payload.**Step 1:
- **Local decision: if soil_moisture < 30% → trigger relay immediately (edge, no cloud latency).**Step 2:
- **ESP32 publishes to RPi MQTT broker over Wi-Fi. RPi aggregates multi-zone readings.**Step 3:
- **RPi forwards data to AWS IoT Core via MQTT/TLS over broadband/cellular.**Step 4:
- **AWS Lambda processes data; SageMaker ML predicts next-day irrigation need from soil + weather forecast.**Step 5:
- **Grafana dashboard shows real-time zone maps; mobile alert for anomalies; agronomist can override pump.**Step 6:
- **App command → MQTT → RPi → ESP32 → relay → pump and solenoid valves activated.**Step 7:

— OR —

(a) WSN in Industrial IoT

In a manufacturing plant WSN, sensor nodes are attached to motors, bearings, and pipes — measuring vibration, temperature, pressure, and power consumption without costly wired runs.

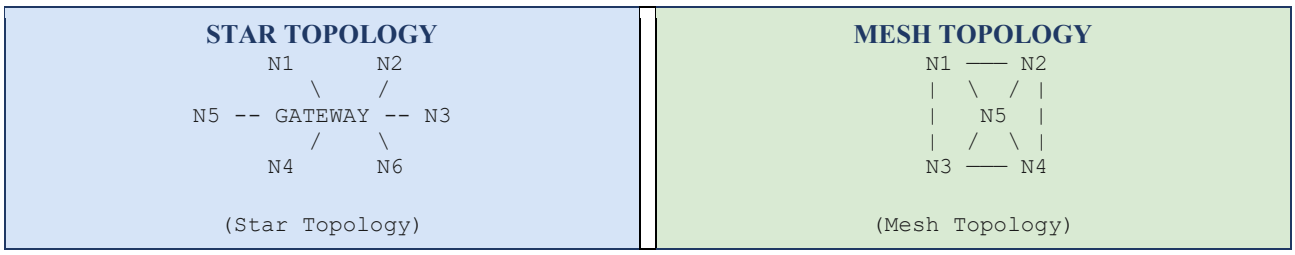

Figure 25: Industrial WSN Topology — Mesh topology provides fault tolerance in factory environments with RF obstructions

- **WirelessHART (IEC 62591), ISA100.11a — backward compatible with existing HART instruments.**Protocols:
- **Mesh — self-healing when nodes fail or RF path is blocked by machinery.**Topology:
- **Predictive maintenance (bearing vibration signatures), leak detection (pressure drop), energy management, gas leak alerts.**Applications:
- **RF interference from motors/drives; metallic multipath fading; hazardous zone battery replacement; OT network security.**Challenges:

(b) Participatory Sensing — Scale & Privacy

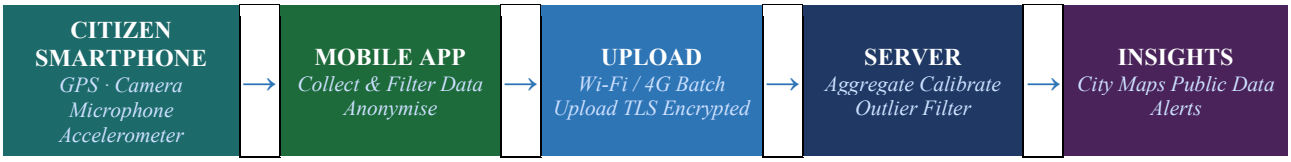

Figure 26: Participatory Sensing — Citizen smartphones as distributed IoT sensor network

- **Waze processes data from 140 million users — equivalent fixed WSN would cost billions to deploy.**Scale advantage:
- **Statistical outlier rejection + calibration against EPA reference monitors.**Quality Control:
- **GPS traces reveal home/work locations, daily patterns, and social connections.**Privacy Concerns:
- **Differential privacy (add calibrated noise), k-anonymity, temporal blurring, on-device ML (upload only inferred results), explicit GDPR-compliant opt-in.**Privacy Solutions: